GDG on Campus
CIT Chennai

Hack Build'26

PITCH DECK TEMPLATE

SkillPulse AI

"Today's Skills. Tomorrow's Workforce."

AI-Powered Labour Market Intelligence & Future Skill Demand-Supply Forecasting Platform

TEAM NAME

Team SkillPulse

CIT Chennai

TEAM MEMBERS(DEPT & SEC/Reg No)

- **S. Dharshan (B.Tech AI & DS / Lead)**
- **Prajan S. (B.Tech AI & DS)**
- Dept of AI & DS, CIT Chennai

HACKATHON TRACK

AI / ML & Predictive Analytics

Open Innovation & Public Policy

Problem Statement

What is the specific problem and who exactly faces it?

- Critical mismatch between educational training output and actual industry skill demand across India.
- Key stakeholders affected: Ministry of Skill Development (MSDE), State Skill Councils, Universities, and Vocational Institutes.
- Curriculums and capacity planning fail to track rapidly emerging sector demands (AI, Data, Cloud, EV, Clean Tech).

What are the critical limitations of existing solutions?

- Static & Outdated Reports: Government surveys and census data take 12·24 months to compile, rendering insights obsolete.
- Purely Retrospective: Existing job boards show only historical postings; they lack quantitative multi-year predictive modeling.
- No Actionable Simulation: Policymakers cannot test 'what-if' seat subsidies to evaluate deficit closure before funding.

Provide brief evidence or a short use case validating the problem.

- National Employability surveys indicate that under 45% of 1.5M+ annual engineering graduates are directly job-ready.
- Simultaneously, specialized roles in AI, Data Engineering, and Cybersecurity take 90+ days to fill due to acute talent shortages.
- State training centers often over-train for saturated legacy trades while emerging high-wage roles remain severely unstaffed.

Proposed Solution

A concise description of how your product solves the problem.

- SkillPulse AI is an AI-powered labour market intelligence platform built for India's Ministry of Skill Development (MSDE).
- It continuously ingests real-time hiring signals, forecasts future demand (12·60 months), and quantifies supply gaps.
- Empowers policymakers and educators with actionable, data-backed seat allocation and training recommendations.

Highlight the top 2-3 essential features.

- Engine 1 · Real-Time Job Intelligence: Ingests 5,000+ jobs across 16 sectors and 13 states with multi-filter search and salary trends.
- Engine 2 · Skills Demand Forecasting: ARIMA baseline + XGBoost predictive models (12, 36, 60 months) with 95% confidence intervals.
- Engine 3 · Decision Policy Simulator: Interactive sandbox adjusting training seats, duration, and completion rates to simulate deficit reduction.

Briefly explain how the target user will interact with the solution.

- Policymakers or university deans select their target sector and state/region on the modern Next.js web application.
- They inspect real-time gap severity levels (Critical, High, Moderate, Surplus) and review emerging skill radar velocity ranks.
- They adjust simulation sliders to immediately calculate the exact training capacity and budget required to achieve balance.

Innovation & Uniqueness

What is the defining characteristic that sets your solution apart?

- Closed-Loop Decision Actionability: Bridges the gap from raw hiring data to executable vocational intervention policies.
- Foresight vs. Hindsight: Transforms workforce planning from backward-looking reports to proactive 1·5 year predictive intelligence.
- Granular Mapping: Connects macroeconomic trends down to 80 specific skill classifications across 5 distinct Indian geographical clusters.

How is this fundamentally better or more effective than the alternatives?

- Uncertainty Quantification: Provides clear upper/lower confidence bounds rather than misleading single-point estimates.
- Ethical & Transparent AI: Strictly segregates empirical historical data, synthetic baselines, and model-derived projections.
- Evidence-Backed ROI: Every recommendation provides a structured evidence summary and projected return on training investment.

Mention any novel methods or technologies used.

- Hybrid Forecasting Pipeline: Integrates statistical ARIMA baseline models with XGBoost regression and residual variance modeling.
- Dynamic Training Absorption Model: Computes batch graduation cadences, dropout rates, and employability lag in the simulator.
- Full-Stack Performance: Built with FastAPI, SQLite/PostgreSQL, SQLAlchemy ORM, and Next.js 14 App Router.

Technical Approach & Feasibility

System Architecture / Workflow Diagram

- Data Ingestion: Schema-validated CSV upload + historical importers -> SQLAlchemy ORM -> SQLite / PostgreSQL.
- Tri-Engine Analytics: Modular Python services for Current Job Intelligence, Time-Series Forecasting, and Gap Simulation.
- Frontend Delivery: High-concurrency FastAPI REST API -> Next.js 14 Web Portal with Recharts interactive visualizations.

Highlight crucial components like AI/ML or specialized APIs.

- Machine Learning Services: Multi-horizon forecasting (12, 36, 60 mo) with MAE, RMSE, and MAPE model evaluation metrics.
- Emerging Skills Radar: Velocity and acceleration ranking algorithms calculating month-over-month growth trajectories.
- REST API Architecture: 15+ endpoints covering /api/dashboard, /api/forecast, /api/gaps/simulate, /api/jobs, and /api/data.

Summarize how you are addressing technical or security hurdles.

- Data Validation & Quality Scoring: Automated schema verification and quality indices prevent ingestion of corrupted records.
- Model Reliability Guardrails: Automatically flags confidence warnings on long horizons (>12 months) or sparse data points.
- Privacy & Compliance: Adheres strictly to open data protocols, stateless API design, environment isolation, and no unauthorized scraping.

Prototype & Demo

Live Demo or GitHub repository link.

- GitHub Repository: https://github.com/luffy-008-sudo/skillpulse-ai- (Full source code, API docs, and models).
- Frontend Web Application: Running locally at http://localhost:3000 (Next.js 14 App Router, modern minimalist design).
- Backend REST API & Swagger: Running at http://localhost:8000/docs (Interactive OpenAPI documentation with live testing).

What core features are successfully implemented right now?

- Engine 1 Active: Searchable catalog of 5,000+ job postings across 16 sectors & 13 states with salary distributions and top skills.
- Engine 2 Active: Time-series forecasting for 12, 36, and 60 months with 95% confidence intervals and Emerging Skills Radar.
- Engine 3 Active: 80 skill-gap analyses across 5 regions + live Training Policy Simulator with adjustable parameters.

A brief note on current progress and visual proof (screenshots).

- Fully Functional Prototype: Complete end-to-end integration between FastAPI backend and Next.js frontend.
- 7 Dedicated Views Built: Market Overview (/), Job Intelligence (/jobs), Forecasts (/forecast), Gap Analysis (/gaps),
- Training Policy Simulator (/simulator), Emerging Skills Radar (/emerging), and Model Performance Evaluation (/models).

Impact, Scalability & Roadmap

Who are the target users and what are the immediate benefits?

- MSDE & State Skill Councils: Prevent severe talent shortages and optimize government skilling fund distribution with data.
- Universities & Technical Institutes: Modernize curriculum roadmaps 2·3 years ahead of industry hiring shifts.
- Students & Job Seekers: Identify high-demand, future-proof career paths backed by transparent salary and hiring indicators.

How will the project grow and sustain itself (business model)?

- B2G & Institutional Subscriptions: Enterprise SaaS licenses for state education boards, vocational bodies, and universities.
- Labour Intelligence API: Tiered API access for recruitment firms, HR analytics providers, and economic research institutes.
- Advisory & Custom Forecasting: Specialized workforce analytics and transition planning for large industrial sectors.

Top 2-3 planned improvements for the future.

- 1. Live Data Feeds: Direct API connectors for Adzuna India, National Career Service (NCS), and e-Shram database.
- 2. Cloud Infrastructure: Migration to enterprise PostgreSQL on AWS/GCP with scheduled worker queues (Celery/APScheduler).
- 3. Hyper-Local Granularity: Expanding forecasting resolution from state level down to 700+ administrative districts.

GDG on Campus
CIT Chennai

Hack Build’26

Thank You!

Innovate Together. Build What’s Next.

TEAM

Team SkillPulse

Built for MSDE & India's Future Workforce

CIT Chennai · Hack Build'26

CONTACT

dharshanshankar08@gmail.com

github.com/luffy-008-sudo/skillpulse-ai-

Active Prototype: localhost:3000